

Prelab report 3 - DSP

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Lab 3: Sampling and reconstruction

Part 1 : Preparation Questions

1) Sampling process

1. The sampling process is the process of taking measurements of a given system in a rate that allows sufficient rate of data loss.

A good example can be the Frames Per Second in computer/tv displays. Most common displays display at around 60hz because the sampling rate of the human eye is around that, give or take 2-3hz. this allows efficient design of displays and programs.

2.
 - The expression for the discrete time fourier transform (DFT) :

$$X(e^{j\theta}) = \sum_{n=0}^{\infty} x[n] \cdot e^{-j\theta n}$$

- The expression for the inverse discrete time fourier transform (IDFT) :

$$x[n] = \frac{1}{2\pi} \int_{\mathbb{R}} X(e^{j\theta}) \cdot e^{j\theta n} d\theta$$

Proof :

We show that the two expressions are the inverse of each other by substituting them in the other expressions and show that they simplify to the original expression.

- We substitute the IDFT definition into the DFT definition :

$$\begin{aligned} x[n] &= \frac{1}{2\pi} \int_{\mathbb{R}} X(e^{j\theta}) \cdot e^{j\theta n} d\theta = \frac{1}{2\pi} \int_{\mathbb{R}} \left(\sum_{m=0}^{\infty} x[m] \cdot e^{-j\theta m} \right) \cdot e^{j\theta n} d\theta = \\ &= \frac{1}{2\pi} \sum_{m=0}^{\infty} x[m] \left(\int_{\mathbb{R}} e^{-j\theta m} \cdot e^{j\theta n} d\theta \right) = \frac{1}{2\pi} \sum_{m=0}^{\infty} x[m] \left(\int_{\mathbb{R}} e^{-j\theta(m-n)} d\theta \right) = \\ &= \frac{1}{2\pi} \sum_{m=0}^{\infty} x[m] \cdot 2\pi \cdot \delta(m-n) = x[n] \end{aligned}$$

- Assuming that all sums and integrals converge, we get back to the former expression of the signal in discrete time domain.

3. The direct DFT does not necessarily exist for every time domain signal $x[n]$. A sufficient condition for which it does exist is the condition of absolute summability of the argument in the definition of the DFT :

$$\sum_{n=0}^{\infty} |x[n]| < \infty$$

4. The relationship between the DFT and the continuous FT is that the DFT is actually the FT sampled at equal distance intervals in the time domain :

$$X(e^{j\theta}) = \frac{1}{T_s} \sum_{n \in \mathbb{Z}} X_c \left(j \left(\frac{\omega}{T_s} - \frac{2\pi k}{T_s} \right) \right)$$

Proof :

We will sample a continuous signal $x(t)$ by multiplying it with an impulse train (dirac delta functions...) $p(t) = \sum_{n \in \mathbb{Z}} \delta(t - n \cdot T_s)$ with constant time intervals with duration T_s :

- We begin with sampling the continuous signal with the pulse train :

$$x_s(t) = x(t) \cdot p(t) = \sum_{n \in \mathbb{Z}} x(t) \cdot \delta(t - n \cdot T_s) = \sum_{n \in \mathbb{Z}} x(n \cdot T_s) \cdot \delta(t - n \cdot T_s)$$

- We take the continuous time FT of the signal :

$$\begin{aligned} X_s(j\Omega) &= \mathcal{F} \left\{ \sum_{n \in \mathbb{Z}} x(n \cdot T_s) \cdot \delta(t - n \cdot T_s) \right\} = \sum_{n \in \mathbb{Z}} x(n \cdot T_s) \cdot \mathcal{F} \{ \delta(t - n \cdot T_s) \} = \\ &= \sum_{n \in \mathbb{Z}} x(n \cdot T_s) \cdot e^{-j\Omega n T_s} = \dots \end{aligned}$$

- We use our notation of the sampled signal $x[n] := x(n \cdot T_s)$ and the notation of the definition of discrete time frequency $\omega := \Omega \cdot T_s$ and notice that the term answers to the definition of the DFT of the sampled signal :

$$\dots = \sum_{n \in \mathbb{Z}} x[n] \cdot e^{-j\omega n} \implies X_s(j\Omega) = X(e^{j\omega T_s}) = \sum_{n \in \mathbb{Z}} x[n] \cdot e^{-j\omega n}$$

- To show the relationship between said signal and the original signal we will use the property of the convolution of the CTFT :

$$\begin{aligned} X_s(j\Omega) &= \frac{1}{2\pi} \cdot X_c(j\Omega) * P(j\Omega) : P(j\Omega) = \frac{2\pi}{T_s} \sum_{k \in \mathbb{Z}} \delta(\Omega - \frac{2\pi k}{T_s}) \\ \implies X_s(j\Omega) &= \frac{1}{T_s} \sum_{k \in \mathbb{Z}} X_c(j\Omega) * \delta(\Omega - \frac{2\pi k}{T_s}) = \frac{1}{T_s} \sum_{k \in \mathbb{Z}} X_c \left(j \left(\Omega - \frac{2\pi k}{T_s} \right) \right) = \end{aligned}$$

- By substituting the definition of the discrete time frequency $\omega := \Omega \cdot T_s$ We get the relationship between the original signal in freq. domain and the DFT of the signal :

$$\implies X(e^{j\theta}) = \frac{1}{T_s} \sum_{k \in \mathbb{Z}} X_c \left(j \left(\frac{\omega}{T_s} - \frac{2\pi k}{T_s} \right) \right)$$

5. We will use matlab to plot the relationship between the 2 signals in Freq. Domain :

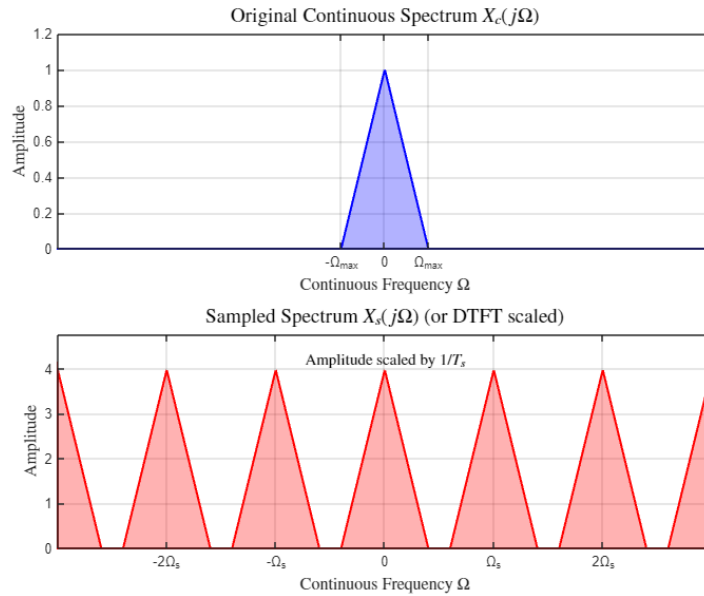


Figure 1: Relationship between the CTFT and the DFT of the sampled signal.
The continuous time signal is being sampled in 2.5 times the original's bandwidth.

6. The relationship between **discrete freq.** and **continuous freq.** is that the continuous freq. bandwidth of the is mapped to the range of $[-\pi, \pi]$ that symbolizes the normalized discrete freq. domain.
Therefore, the difference is that the discrete freq. domain of a function assumes that the signal was originally sampled at Nyquist freq. and holds all original data of the signal upto that max freq.
7. Using the relationship we derived in 1.4, we can see that the term being summed has bandwidth identical to the original's signal bandwidth and is duplicate in equal intervals of $\frac{2\pi}{T_s}$.
Therefore, if we want to prevent aliasing we need to move the duplicates of the signal in freq. domain, far enough from each other by increasing T_s .
8. No, since aliasing yields a signal that can be possibly reconstructed in infinite number of ways. We want no aliasing so we can reassure that the values in freq. domain are the output of a single function and then we can be sure of its reconstruction.
9. Since the original signal doesn't have a bounded bandwidth, we want to achieve 2 goals :
 - Have no aliasing to allow a legit reconstruction.
 - To reconstruct at least 98% of the original signals power.

Solution:

- To tackle the first goal we want to cut the signal at some freq. to make the reconstruction possible. Since the signal's bandwidth is unbounded will set a cutoff freq. for the LPF we will use to cut the signal.
- To tackle the second goal we will choose the cutoff freq. such that at least 98% of the signal's power is in the bandwidth.

- After choosing the cutoff freq. accordingly, we will first pass the signal with the LPF, send it over the communication channel and then performs a reconstruction of the signal.
 - $\implies$ We will have a sufficient amount of the original's signal bandwidth reconstructed without aliasing!
10. We will derive the expression for the difference between the original signal and reconstructed signal and evaluate the error for both cases :
- **We use anti-aliasing filter :**
We will have sufficient spacing between duplicates of the bandwidth and will not have any error around the origin :

$$E_1(j\Omega) = X_c(j\Omega) - X_{r,1}(j\Omega) = \begin{cases} 0 & \text{for } |\Omega| < \Omega_c \\ X_c(j\Omega) & \text{for } |\Omega| \geq \Omega_c \end{cases}$$

- **We Don't use anti-aliasing filter :**
The "tails" of other duplicates will affect freq. inside the baseband :

$$E_2(j\Omega) = X_c(j\Omega) - X_{r,2}(j\Omega) = \begin{cases} -\sum_{k \neq 0} X_c(j(\Omega - k\Omega_s)) & \text{for } |\Omega| < \Omega_c \\ X_c(j\Omega) & \text{for } |\Omega| \geq \Omega_c \end{cases}$$

11. The error from the first case stems from the truncation. We intentionally give up a part of the signals information to prevent further loss due to aliasing.
12. The error from the second case stems from the aliasing. We get a signal that is comprised of 2 sources and therefore there is not absolute method of reconstruction and therefore we declare the information lost.

2) Quantization

1. Quantization is the process of collecting data with finite precision. In the context of the sampling process, the quantization comes in 2 forms :
- **Time domain x-axis :** The sampling process has a finite accuracy since the sampling rate is finite and since the data stored in memory is discrete in contrast to real world signals that are continuous.
 - **Amplitude y-axis :** The precision of data structure holding the information about the amplitude is finite. Therefore, we round the numbers to the closest precision point and lose some of the information to some extent.

2. We will explain the effect of the Quantization for both cases :

- **Number of Bits (B):** The number of bits determines the total number of available quantization levels (2^B). As the number of bits increases, the quantization step size (Δ) decreases. Since the quantization error is typically bounded within the range of $[-\frac{\Delta}{2}, \frac{\Delta}{2}]$, increasing the number of bits directly shrinks the absolute domain of the error.
- **Representation of Negative Numbers:** Methods like "Sign-Magnitude" and "1's Complement" have two representations for zero ($+0$ and -0), which wastes one valuable quantization level. In contrast, using "2's Complement" utilizes all levels efficiently without a redundant zero, allows for a slightly wider dynamic range (representing one additional negative number), and prevents extra computational errors around zero.
- **Fixed-point vs. Floating-point:**
 - **Fixed-point:** In this representation, the quantization step size (Δ) is constant across the entire range. Therefore, the absolute quantization error domain remains strictly constant and is completely independent of the sampled signal's amplitude.
 - **Floating-point:** The system represents numbers using a mantissa and an exponent, meaning the step size (Δ) scales relative to the signal's magnitude. For small signals, the absolute error is small; for large signals, the absolute error is large. This means the absolute error domain changes dynamically, but the *relative* error (the error relative to the signal's size) remains approximately constant.

3. We plotted methods of Quantization in matlab :

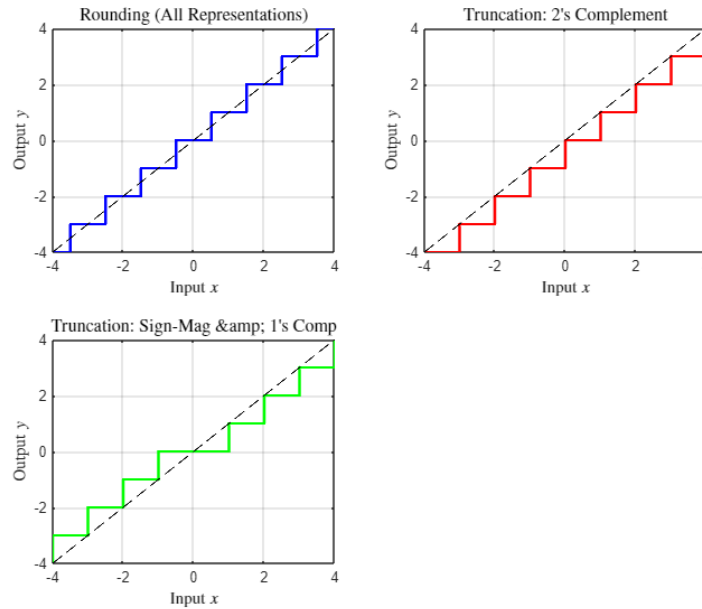


Figure 2: representations of different methods of Quantization

Each of the plots was received using the following function:

- **Rounding** : $y = \Delta \cdot \text{Round}(\frac{x}{\Delta})$
- **Truncation 1's Complement** : $y = \Delta \cdot \lfloor \frac{x}{\Delta} \rfloor$
- **Truncation 2's Complement** : $y = \Delta \cdot \lfloor \frac{|x|}{\Delta} \rfloor$

4. See previous subsection.

5. We are presented with the following statistical model for quantization of an analog signal:

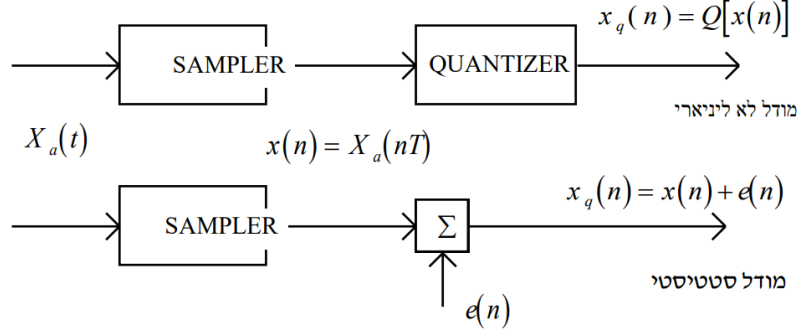


Figure 3: representations of different methods of Quantization

(a) The statistical assumptions for quantization model are as such :

- The distribution of the quantization error is uniform over all step sizes at distance Δ from each other.
- The quantization error samples are modeled as white noise, meaning that any 2 different samples of the error are statistically independent of each other : $e[n] \perp e[m] : \forall n \neq m$
- The quantization errors are uncorrelated with the data samples : $e[n] \perp x[m] : \forall n, m \in \mathbb{Z}$

(b) **Example** : Imagine a constant DC input signal, such as $x(n) = 3.2$ Volts for all n . If our quantizer rounds this to the nearest integer, the quantized output is $x_q(n) = 3.0$ Volts for all n . The error is therefore $e(n) = -0.2$ for every single sample.

- It is not uniformly distributed (it is a single constant value).
- It is not white noise.
- It is 100% correlated with (and entirely determined by) the input signal $x(n)$.

(c) Assuming the statistical model holds, the error follows a Uniform Probability Density Function (PDF), denoted as $f_e(e)$. The only difference between rounding and truncation is where this uniform distribution is centered. Let Δ be the quantization step size.

- **Rounding** : The error is evenly spread around zero, up to half a step size in either direction.

$$f_e(e) = \begin{cases} \frac{1}{\Delta} & \text{for } -\frac{\Delta}{2} \leq e \leq \frac{\Delta}{2} \\ 0 & \text{otherwise} \end{cases}$$

- **Truncation** : Assuming standard 2's complement truncation (which always rounds downwards towards $-\infty$), the quantized value is always less than or equal to the real value. Therefore, the error $e(n) = x_q(n) - x(n)$ is always negative, spread evenly between $-\Delta$ and 0.

$$f_e(e) = \begin{cases} \frac{1}{\Delta} & \text{for } -\Delta \leq e \leq 0 \\ 0 & \text{otherwise} \end{cases}$$

- (d) For a uniform quantizer (assuming a full-scale sinusoidal input signal and the statistical noise model we discussed in 2.5), the theoretical Signal-to-Noise Ratio expressed in decibels (dB) is given by the famous formula:

$$SNR_{dB} = 10 \log_{10}(1.5) + 20B \log_{10}(2) \approx 6.02B + 1.76 \text{ dB}$$

- (e) To maximize the Signal-to-Noise Ratio (SNR) and prevent distortion. You want the signal to use as many quantization levels as possible without exceeding the hardware's physical voltage limits. What happens in each specific case?

- If $|V_{inMax}| > |V_{A/DMax}|$ (Clipping): The peaks of the signal are cut off by the hardware limits, causing non-linear distortion.
- If $|V_{inMax}| \gg |V_{A/DMax}|$ (Severe Clipping): The signal is completely flattened against the limits, turning into a square wave and destroying the original waveform shape.
- If $|V_{inMax}| < |V_{A/DMax}|$ (Reduced SNR): The signal "wastes" available quantization levels. The fixed quantization noise becomes a larger percentage of your signal, lowering the SNR.
- If $|V_{inMax}| \ll |V_{A/DMax}|$ (Severe Noise Distortion): The signal is so small it spans only a few discrete levels. It gets entirely buried in quantization noise and looks like a blocky staircase.

3) Reconstruction

1. Here is the formula for **Shannon's interpolation** :

$$X_a(t) = \sum_{n \in \mathbb{Z}} x(n \cdot T_s) \cdot \text{sinc}\left(\frac{t - n \cdot T}{T}\right) : \left\{ \text{sinc}(x) := \frac{\sin(\pi x)}{\pi x} \right\}$$

We can see that when substituting $X_a(t)|_{t=n \cdot T}$ all every pulse in the interpolation cancels out except for the one in the origin.

Proof :

- Since we are dealing with a band-limited signal, we can treat it as if we passed our signal through an LPF with cutoff freq. that matches the Nyquist sampling rate as follows

$$\begin{aligned} h(t) &= \frac{1}{2\pi} \int_{-\pi/T}^{\pi/T} T e^{j\Omega t} d\Omega = \frac{T}{2\pi} \left[\frac{e^{j\Omega t}}{jt} \right]_{-\pi/T}^{\pi/T} = \frac{T}{\pi t} \left(\frac{e^{j\frac{\pi}{T}t} - e^{-j\frac{\pi}{T}t}}{2j} \right) = \\ &= \frac{T}{\pi t} \sin\left(\frac{\pi}{T}t\right) = \frac{\sin(\pi \frac{t}{T})}{\pi \frac{t}{T}} = \text{sinc}\left(\frac{t}{T}\right) \end{aligned}$$

When $h(t)$ is the impulse response of the LPF in time domain.

- Now we can represent the the analog signal as a convolution of the original signal with the LPF :

$$X_a(t) = x_s(t) * h(t) = \int_{-\infty}^{\infty} \left(\sum_{n=-\infty}^{\infty} x(n) \delta(\tau - nT) \right) h(t - \tau) d\tau$$

- We get the final expression after substitutions and using dirac's delta identity :

$$X_a(t) = \sum_{n=-\infty}^{\infty} x(n) \text{sinc} \left(\frac{t - nT}{T} \right)$$

2. For a band limited signal, the claim always holds. For a non-band limited signal, the claim only holds if the signal is samples at exactly $t = n \cdot T : n \in \mathbb{Z}$.
3. No, since the Shannon filter is not causal and has infinite impulse response in time domain.
4. A standard, real-world example of a reconstructor that solves the issues of the ideal Shannon formula is the Zero-Order Hold (ZOH) , combined with a practical analog smoothing filter. Here is how it overcomes the two specific problems from 3.3:
 - (a) It is Causal : The ZOH reconstructor does not need to look into the future. When a sample $x(n)$ arrives, the ZOH simply outputs that exact voltage level and holds it constant until the next sample arrives.
 - (b) It has Finite Duration : Instead of an infinite sinc function, the impulse response of a ZOH is just a simple rectangular pulse of width T (the sampling period). It requires virtually no memory and processes instantly.